

A novel approach to trading strategy parameter optimization, using double out-of-sample data and walk-forward techniques*

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Abstract

This study introduces a novel approach to walk-forward optimization by parameterizing the lengths of training and testing windows. We demonstrate that the performance of a trading strategy using the Exponential Moving Average (EMA) evaluated within a walk-forward procedure based on the Robust Sharpe Ratio is highly dependent on the chosen window size. We investigated the strategy on intraday Bitcoin data at six frequencies (1 minute to 60 minutes) using 81 combinations of walk-forward window lengths (1 day to 28 days) over a 19-month training period. The two best-performing parameter sets from the training data were applied to a 21-month out-of-sample testing period to ensure data independence. The strategy was only executed once during the testing period. To further validate the framework, strategy parameters estimated on Bitcoin were applied to Binance Coin and Ethereum. Our results suggest the robustness of our custom approach. In the training period for Bitcoin, all combinations of walk-forward windows outperformed a Buy-and-Hold strategy. During the testing period, the strategy performed similarly to Buy-and-Hold but with lower drawdown and a higher Information Ratio. Similar results were observed for Binance Coin and Ethereum. The real strength was demonstrated when a portfolio combining Buy-and-Hold with our strategies outperformed all individual strategies and Buy-and-Hold alone, achieving the highest overall performance and a 50% reduction in drawdown. A conservative fee of 0.1% per transaction was included in all calculations. A cost sensitivity analysis was performed as a sanity check, revealing that the strategy's break-even point was around 0.4% per transaction. This research highlights the importance of optimizing walk-forward window lengths and emphasizing the value of single-time out-of-sample testing for reliable strategy evaluation.

Keywords: Testing Architecture, Optimization Techniques, Cryptocurrencies, Moving Averages, Algorithmic Trading, Intraday Trading, High Frequency Trading, Bitcoin, Walk-Forward trading, Rolling window optimization

JEL Codes: C4, C14, C45, C53, C58, G13

*The associated GitHub repository covering all calculations presented in this paper can be found at: https://github.com/tmr-crypto/wf_optim_crypto_analysis

1 Introduction

The development of trading strategies in stock and cryptocurrency markets has been extensively explored through technical analysis in prior research. However, these studies frequently overlook critical elements such as transaction fees, slippage, and other trading costs. Additionally, a common but often unreported issue involves repeated optimizations on data designated as out-of-sample, or the complete omission of an out-of-sample evaluation phase. Such practices foster unrealistic performance expectations, leading to substantial underperformance when strategies are implemented in live trading environments.

This study addresses key challenges in trading strategy development by providing a practical framework for constructing and assessing trading strategies. The framework incorporates tools for profitability estimation, statistical significance evaluation, and overfitting risk mitigation. Central to this investigation is the integration of the Exponential Moving Averages (EMA) Crossover—a momentum-based technical analysis strategy—with the walk-forward optimization procedure, which serves as the foundational structure of our approach.

The study investigates the following research questions:

RH1: Walk-forward optimization, a method widely used in this work, divides the research period into training/testing subperiods instead of optimizing parameters over the entire dataset, which risks overfitting. For each training subperiod, optimal parameters are identified and evaluated out-of-sample in the corresponding testing subperiod. Out-of-sample results across all testing periods are aggregated to assess overall performance. This study examines the effects of varying training and testing window lengths on risk-adjusted returns, aiming to identify any dependence between window length and investor outcomes. It explores whether shorter or longer windows influence results, contrasting with the Efficient Market Hypothesis (Fama 1965), which suggests market efficiency precludes significant performance differences from window length variations.

RH2: How do the choice of data frequency and costs impact strategy results? This research utilizes intraday high-frequency data, with frequencies ranging from 1 to 60 minutes evaluated. The paper assesses the influence of costs and time frequency on risk-adjusted returns.

RH3: Is the tested strategy better than random? To evaluate statistical significance, the bootstrap methodology is applied to determine if the optimal strategy outperforms randomly constructed alternatives generated during simulations.

Walk-forward optimization, an established technique in time series analysis, is employed to avoid overfitting in parameter estimation. This study extends beyond conventional fixed window lengths by examining the effects of varying training and testing window lengths on strategy

performance, potentially enhancing strategy efficacy and contributing to a broader theoretical understanding of walk-forward optimization. Data utilization is approached innovatively by partitioning available data into a global training period for parameter optimization and an unseen period for execution, thereby reducing data mining bias. This differs from typical practices where strategies are iteratively tested on purported out-of-sample data, compromising their integrity. The analysis incorporates intraday cryptocurrency data, including Bitcoin, Ethereum, and Binance Coin, with parameters derived from Bitcoin training applied across assets to test adaptability.

A novel aspect of this work is the systematic investigation of training and testing window lengths in walk-forward optimization, moving away from arbitrary fixed lengths to uncover their impact on performance. Results indicate that the EMA Crossover strategy combined with walk-forward optimization reduces volatility and drawdown relative to Buy-and-Hold approaches. When integrated into portfolios with Buy-and-Hold, it enhances all performance metrics. The strategy's parameters, optimized on Bitcoin, yield superior results when applied to Ethereum and Binance Coin in the testing period. Furthermore, constructing portfolios blending passive Buy-and-Hold with the active EMA-walk-forward strategy reveals untapped diversification benefits, achieving optimal risk-adjusted measures and drawdown reductions compared to standalone Buy-and-Hold or walk-forward strategies. Overall, the framework demonstrates adaptability to evolving market regimes, with the strategy delivering positive returns for nearly two years on intraday cryptocurrency data without retraining. This persistence prompts further inquiry into potential improvements via periodic retraining during testing.

The paper is organized as follows: Section 2 presents a review of relevant academic literature. Section 3 details the methodology employed. Subsequent sections report the results, including analyses of cost sensitivity and seasonality in strategy returns. Finally, results on unseen data are discussed, followed by conclusions.

2 Literature Review

Technical analysis remains underrecognized in academia, likely due to its simplicity and lack of alignment with financial ratios or economic indicators. Despite limited scientific explanations, a Group of Thirty (1985) survey found that 97

Park and Irwin (2007) reviewed 92 studies on technical analysis, showing 58 profitable, 24 negative, and 10 mixed results. Pre-1980s strategies were profitable in futures and foreign exchange but not in stocks. A recent review by Nti et al. (2020) of 122 papers on technical and

fundamental analysis reported 87

Hsu et al. (2010) noted that ETFs may reduce technical strategy profitability, yet inefficiencies persist in immature markets, attracting arbitrageurs and enhancing efficiency. Challenges include data limitations and assumptions of closing-price executions. Zielonka (2004) found investors overestimate technical signals due to cognitive biases like confirmation bias, favoring aligned beliefs; simple rules like moving average crossover are often neglected.

Neftçi (1991) described moving average crossover as statistically well-defined, unlike most technical rules. Farias Nazário et al. (2017) identified it as the most popular method (44 of 80 studies), surpassing neural networks or genetic algorithms (15 each). Hybrid approaches include Ayala et al. (2021), combining TEMA and MACD with SVMs and Random Forests for positive results on stock indices, excluding costs. Karasu et al. (2020) integrated EMA, SMA, Kaufman’s adaptive moving average, MOPSO, and RBFSVR to improve crude oil price forecasting over traditional methods.

Brock et al. (1992) tested moving average crossover, yielding higher returns and lower volatility post-buy signals and negative returns with higher volatility post-sell, outperforming AR(1), GARCH, and random walk models. Zakamulin and Giner (2020) compared momentum and moving average crossover, finding similarities in strong trends; under random walk, they equate to coin flips, but with trends or mean reversion, they exceed 50

As a recent phenomenon, cryptocurrencies have drawn growing academic interest with mixed results. Brown and Pelosi (2019) applied moving average crossover to 40 cryptocurrencies, outperforming Buy-and-Hold in only 10; 10 lacked data, and Buy-and-Hold won in the remaining 20.

Walk-forward analysis divides time series into sequential in-sample and out-of-sample periods, optimizing parameters in-sample and evaluating out-of-sample. Pardo (2012) argued it reduces overfitting, boosts reliability, and assesses market adaptation, though it requires substantial data and computation. López de Prado (2019) favored Monte Carlo with synthetic data over walk-forward, as the latter assumes consistent data processes and ignores stochasticity. LeBaron and Weigend (1998) critiqued train/validation/test splits for overestimating performance, using bootstrapping to show variability from splits, not parameters.

Karathanasopoulos et al. (2016) used 300- and 400-day sliding windows to predict crude oil-refined product spreads, finding 300-day windows with particle swarm optimization and radial basis function neural networks slightly superior in risk-adjusted returns. In cryptocurrencies, Ahmed

et al. (2020) applied moving average crossover to Dash (2016-2018), yielding 14-18% excess over Buy-and-Hold, but not for a 10-coin privacy portfolio. Wen et al. (2022) identified intraday predictability, momentum, and reversal in cryptocurrencies, impacted by events like FOMC meetings and COVID-19; strategies outperformed Buy-and-Hold, excluding costs.

Alonso-Monsalve et al. (2020) used neural networks with one-minute data and technical indicators to predict cryptocurrency returns, omitting costs. Wei (2018) found no illiquidity premium in 456 cryptocurrencies, with illiquid ones showing antipersistence (low Hurst exponents) and deviations from randomness, unlike efficient liquid markets. Anghel (2021) applied White’s Reality Check to technical and machine learning strategies, finding rare significant returns after data snooping and frictions; machine learning underperformed technical analysis due to higher transactions, with no added benefits over traditional markets. Kosciuszko et al. (2019) examined momentum and contrarian effects in 100 large-cap cryptocurrencies, with short-term contrarian outperforming momentum and benchmarks, offering diversification over the S&P 500.

Limitations persist: Farias Nazário et al. (2017) noted only 37% of studies adjust for risk. Park and Irwin (2007) highlighted biases like data snooping, rule selection, and cost estimation difficulties. Efron (1979) popularized bootstrapping for i.i.d. data, but financial series violate assumptions via autocorrelation and heteroscedasticity, per Ruiz and Pascual (2002); block bootstrapping by Politis and Romano (1992) addresses this.

Presented research provides insights into technical analysis, though academia lacks consensus on its effectiveness or foundations. This study contributes additional questions to the discussion rather than definitive answers.

3 Data

3.1 Description of data

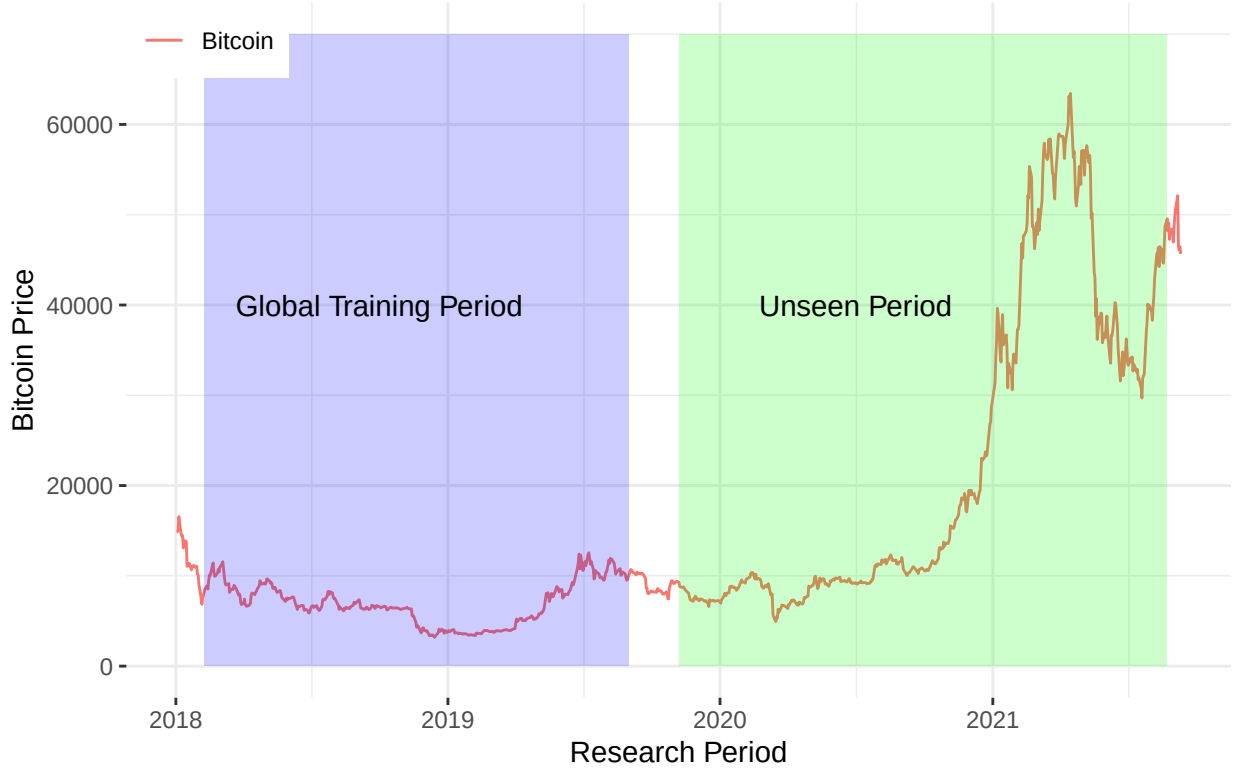
The analyzed time series represent cryptocurrency prices. The data was acquired from Kaggle.com, which hosted a predictive modeling challenge called G-Forecast in collaboration with a commercial partner. The contest data sampled the prices of 10 different cryptocurrencies at one-minute intervals, although only a subset was ultimately used in this research.

The data encompasses two distinct periods. Notably, the Unseen Data Period is slightly longer than the Global Training Data Period:

- Global Training Data Period: February 8, 2018, to September 1, 2019
- Unseen Data Period: November 7, 2019, to August 22, 2021.

The gap between the end of the Global Training Data Period and the start of the Unseen Data Period arises from a walk-forward adjustment to ensure both periods begin and end at the same time, despite potential differences in data frequency.

Figure 1: Price of Bitcoin with marked Global Training/Unseen Period



Note: Bitcoin price (shaded area indicates data division). The global training data period covers data from February 8, 2018, to September 1, 2019. The unseen data period covers data from November 7, 2019, to August 22, 2021.

3.2 Description of the Global Training Data Period

3.3 Basic statistic properties

Table 1 explores the basic statistical properties of the Bitcoin price time series across various sampling intervals.

Key findings:

- While high kurtosis is observed consistently, interestingly, it exhibits a decreasing trend with increasing sampling intervals.
- This pattern is disrupted at the 5-minute interval, where kurtosis remains higher compared to the 1-minute interval. Further investigation is needed to understand the behavior at the 5-minute sampling. High kurtosis indicates a higher probability of large price movement and a leptokurtic feature of time series; findings align with previous studies, such as the work by Petukhina Alla A. and Karl (2021), where the kurtosis of intraday data was

estimated. It is worth noting that the mentioned research has documented kurtosis for Bitcoin as high as 49 for 5-minute data, which is much higher than presented below.

- Across all sampling intervals, the data exhibit positive skew, meaning the right tail extends further than the left. This indicates a higher probability of large positive price movements compared to large negative ones.
- Additionally, the maximum values observed in each sampling interval are consistently larger in absolute terms for positive deviations than for negative deviations, further confirming the asymmetry of the distribution.

Table 1: Descriptive statistics of the global training data for Bitcoin (BTC)

Freq Minutes	Mean	SD	Median	Min	Max	Range	Skew	Kurtosis	JB P-Value
1	0	0.0013	0.0000	-0.0465	0.0315	0.0779	-0.0769	18.13	0
5	0	0.0027	0.0000	-0.0792	0.0589	0.1382	0.0102	31.97	0
10	0	0.0037	0.0000	-0.0760	0.0610	0.1369	-0.0324	26.16	0
15	0	0.0045	0.0000	-0.1364	0.0770	0.2134	-0.3408	37.87	0
30	0	0.0064	0.0000	-0.0878	0.1033	0.1911	0.0775	22.63	0
60	0	0.0089	0.0001	-0.0925	0.1081	0.2006	0.1598	16.08	0

Note: Descriptive statistics of the global training data for Bitcoin (BTC). JB P-Value - Jarque-Bera test p-value. The global training data period covers data from February 8, 2018, to September 1, 2019. All statistics are presented on a non-annualized basis.

4 Methodology

The study employed various methodologies to minimize the influence of cognitive biases in strategy results. To decrease the risk of data mining bias, during the research architecture design phase, the data was divided into two main periods: the Global Training Data Period and the Unseen Data Period. All strategy parameter optimization was performed on the Global Training Data Period, while the Unseen Data Period was used only once to calculate the performance for the final best parameter combination. Using the Unseen Data Period only once ensures that the results reflect the strategy’s genuine performance on truly unseen data and are less susceptible to data mining bias.

A novel approach to strategy optimization was used. Instead of solely finding the best Moving Average crossover parameters, an additional level of optimization involved finding the optimal lengths of walk-forward training/testing periods. In typical studies, the length of walk-forward steps is chosen arbitrarily. However, in this work, it is part of the optimization process. Currently,

the literature covering optimization related to the choice of the window size period is very limited. Although some studies investigate this in terms of sensitivity analysis, for example, Karathanasopoulos et al. (2016) used two different windows for the testing period: 300 days and 400 days. In other research, Kryńska and Ślepaczuk (2023) investigated the effect of varying in-sample and out-of-sample lengths on their trading strategy results. They found that extending the training period (in-sample) while decreasing the testing period (out-of-sample) resulted in improved performance.

The strategy built in this study uses multiple techniques tailored to its specific needs, including:

- Walk-forward method optimization
- Technical analysis indicator Exponential Moving Average (EMA) with optimized length of MA periods.
- Bootstrap

4.1 The Walk-Forward Method

This research employed walk-forward optimization (WFO) to ensure that all reported results came from out-of-sample data - either from WFO testing periods or single-time evaluations on the Unseen Data Period. This approach limits the probability of overfitting, which occurs when strategies optimized on in-sample data fail to generalize to new market conditions.

The WFO method is just an adaptation of a popular technique used in data science and quantitative finance, where the validity and the robustness of model parameters are typically evaluated by dividing the data into different sets:

- Training set (in-sample data): This data is used to train the model and fine-tune its parameters
- Validation set (optional in some cases): This data controls the training process on data that has not been seen by the model
- Testing set (out-of-sample data): This set of data provides the final assessment of the model’s capabilities to generate good performance on truly new data.

The walk-forward optimization technique is a method used to evaluate the robustness and potential profitability of trading strategies. It involves dividing the historical data into n equal periods. Each of these n periods is further divided into two or three when a validation set is included, into segments with different lengths:

- In-sample/training period: This segment is used to optimize the parameters of the trading strategy. Essentially, the strategy is “trained” on this data to find the settings that yield the best performance. In this study, the training period is used to find a combination of

fast/slow lengths of Exponential Moving Average that maximizes Sharpe.

- Out-of-sample/testing period: This segment is used to validate the performance of the strategy using the parameters found in the in-sample period. This acts as a simulation of applying the strategy to the new set of data, helping to avoid overfitting and assess its adaptability to different market conditions. In the context of this study, the out-of-sample period evaluates a combination of strategy parameters - fast/slow of the Exponential Moving Average found in the in-sample period.

Combining results: Only the results from all out-of-sample periods are used to calculate overall statistics for the global training period. This prevents overfitting by avoiding the use of in-sample (training) data in performance evaluation.

The impact of using different training and testing data lengths within the walk-forward optimization process on strategy performance was further investigated and detailed in the following sections.

4.2 Exponential Moving Average Crossover

It is a momentum strategy using moving averages to generate buy and sell signals. The simplest moving average crossover strategy uses two averages of prices. One is shorter and sometimes called fast, while the other one is longer and called slow. Crossover occurs when a short-term moving average crosses through a long-term moving average. When a crossover occurs, practitioners believe the momentum of the price is shifting its direction.

This study employed a set of Exponential Moving Averages (EMAs) with periods of 5, 7, 10, 15, 20, 30, 40, 50, 100, 150, and 200. EMAs with periods longer than 35 were categorized as slow, while those with shorter periods were classified as fast. The selection of this threshold was somewhat arbitrary.

Formula (1) below presents the recursive form of the Exponential Moving Average:

$$EMA_n = \alpha * P_n + (1 - \alpha) * EMA_{n-1} \quad (1)$$

where :

n – Number of periods for which EMA should be calculated

P_n – Current asset price

$$\alpha = \frac{2}{n + 1}$$

$$EMA_0 = P_0$$

4.2.1 Position of strategy

For a specific combination of fast and slow Exponential Moving Average (EMA) parameters, the strategy:

- Enters long positions (buy) when the fast EMA crosses above the slow EMA. This indicates a potential upward trend, suggesting the price might continue to rise, and the strategy aims to capitalize on this movement.
- Enters short positions (sell) when the fast EMA crosses below the slow EMA. This suggests a potential downward trend, and the strategy aims to profit from a price decline.

The logic for position calculation could be summarized in the formula (2):

$$EMA_{\text{Position}} = \begin{cases} \text{Long,} & \text{if } EMA_{\text{fast}} \geq EMA_{\text{slow}} \\ \text{Short,} & \text{otherwise} \end{cases} \quad (2)$$

4.3 Optimization

This study went beyond optimizing EMA parameters and introduced an additional layer of optimization by fine-tuning the training/testing period length for the walk-forward, potentially leading to a more robust strategy.

4.3.1 Specific periods of walk-forward training/testing considered

To investigate the presence of potential weekly seasonality, in addition to shorter periods of 1, 2, 3, 5, 7, and 10 days, the study explored various training/testing WF period combinations using multiples of 7 days. These included periods of 14, 21, and 28 days.

4.3.2 Optimization approach

Each combination of training and testing window sizes underwent a separate walk-forward optimization process. This process used the entire global training period to assess the impact of specific window sizes on the strategy’s performance. Each iteration of the walk-forward procedure with a unique window size combination produced a Sharpe ratio, which was then stored for further analysis. Importantly, the Sharpe Ratio was calculated based on the walk-forward returns, specifically derived from the test phase of each of the walk-forward steps.

4.3.3 Visualization

The overall results of the above calculation can be presented as a grid in Figure 2. The horizontal axis represents the training window size, and the vertical axis represents the test window size. Each intersection point displays the Sharpe ratio achieved using that specific combination for the global training period.

4.3.4 Smoothing grid

The grid presented in Figure 2 reveals clusters of higher and lower Sharpe ratios across different combinations of training and testing periods. However, directly choosing a single set of parameters is challenging due to the high variability in results in neighboring states. To address this variability, a weighted smoothing method was employed. This method weights the Sharpe ratio of a cell based on the average Sharpe ratios of its neighboring cells, effectively giving more weight to areas with consistent positive performance.

4.3.5 Choice of Smoothing Weights

While the specific choice of smoothing weights can be subjective, in this case, they were chosen following the method described in Gómez and Ślepaczuk (2021). Smoothing is achieved by using the formula (3) for a weighted average where the original value receives half the weight, and the remaining weight is distributed among neighboring values.

$$Smoothed_{value} = \frac{1}{2} * Original_{value} + \frac{1}{2} * \sum_{n=1}^n \frac{1}{n} Neighbor_{value} \quad (3)$$

4.3.6 Identifying optimal parameters for out-of-sample evaluation based on the smoothed grid

After performing the smoothing procedure described in the previous paragraph, the two walk-forward parameter combinations with the highest Sharpe ratios in the smoothed grid (Figure 3) are chosen for further analysis. This selection method is objective and based on the chosen performance metric (Sharpe ratio). However, the study acknowledges that this approach might not be the most effective, and alternative methods for selecting optimal walk-forward parameters could be explored. Alternative methods could include:

- Minimizing the difference between training and testing period results. This approach aims to select parameters that lead to similar performance in both training and testing phases, suggesting better generalizability to unseen data.
- Using bootstrapping to select the walk-forward parameter combination with the highest statistical significance. Bootstrapping is a resampling technique that can help assess the reliability of the parameter selection process.

4.3.7 Overall optimization summary

This optimization process effectively assessed the impact of different training and testing period combinations on the strategy's performance, providing valuable insights for selecting a more robust parameter set. However, it is important to acknowledge potential limitations:

- Data mining: Analyzing numerous training/testing window size combinations, combined with selecting the best EMA within the walk-forward training period, raises concerns about data mining. This practice can lead to overfitting, where the chosen parameters perform well on the specific data used but may not generalize well to unseen data.
- Training/unseen data split: The chosen split between the global training and unseen data periods can also influence parameter selection. Different splits could potentially lead to different optimal parameters.
- Parameter decay: The optimization process does not consider the possibility that the chosen parameters might become less effective over time. The strategy's performance on unseen data could potentially deteriorate if the parameters found in the global training period are no longer optimal.

These limitations highlight the need for further investigation into robust parameter selection techniques and the potential for performance decay over time.

4.4 Statistical significance

This study employed bootstrapping to evaluate statistical significance. However, as noted by Ruiz and Pascual (2002), the classical bootstrap introduced by Efron (1979) is not suitable for time series data due to potential heteroscedasticity and autocorrelation. Therefore, the block bootstrap method developed by Politis and Romano (1994) was customized to specifically address these concerns and better fit the nature of time series data, and was the method used in this paper.

4.4.1 Bootstrap Evaluation of Subsampled EMA Combinations for Walk-Forward Optimization

Within each step of the walk-forward training period, the study evaluated various combinations of exponential moving averages (EMAs). The combination that generated the highest Sharpe Ratio during the training period was then chosen for further assessment on the subsequent walk-forward testing period. In contrast, the bootstrapping method employed a different approach. Instead of selecting the best EMA combination based on the training data, it utilized random selection. This effectively compared the performance of the “best” parameter choice, identified during the walk-forward optimization, against a baseline established by randomly chosen combinations. Due to the computationally intensive nature of this method (requiring 1,000 iterations), it was only applied to the top two train/testing period combinations identified through their Sharpe Ratio during the walk-forward process on the global testing period.

4.4.2 Bootstrap Evaluation of Shuffled Transaction blocks

The transactional positions bootstrap was another method of bootstrap that was adapted for this study. During the walk-forward testing period, “position blocks” were generated. These blocks represented consecutive periods where the strategy held long or short positions (e.g., 3 periods long, followed by 2 short, then 5 long). In each bootstrap iteration, the order of these blocks was shuffled. For example, the 2-period short block might be swapped to come before the 8-period long block (5 + 3), and then original asset returns are matched to each block to construct the equity line and calculate profitability for a single iteration. Due to the computationally intensive nature of this method (requiring 1,000 iterations), it was only applied to the top two train/testing period combinations identified through their Sharpe Ratio during the walk-forward process on the global testing period.

4.4.3 Null hypothesis

Both block and moving average (MA) bootstrapping methods assumed a null hypothesis that the research strategy does not provide any statistically significant advantage over randomly generated trading signals. In simpler terms, the hypothesis assumed the strategy's performance is purely due to chance.

To reject this null hypothesis and conclude that the strategy has a statistically significant edge, it needs to outperform a predetermined proportion of random strategies based on the chosen confidence level.

Confidence level represents the probability that the observed effect (strategy's edge) didn't occur by chance. A higher confidence level indicates a stronger likelihood that the effect is real.

For example, with a 5% confidence level, the strategy needs to be demonstrably better than 95% of randomly generated strategies to be considered statistically significant. This implies that the strategy's performance should consistently outperform 95% of randomly generated signals, suggesting it has a genuine advantage over a random approach.

4.5 Cost sensitivity

While the influence of transaction costs on the strategy outcomes has received limited academic attention, this study delved into this subject. One of the previous studies by Svogun and Bazán-Palomino (2022) demonstrated the significant impact of transaction costs on trading strategies using intraday data (1-minute), while having a minimal effect on strategies using 1-day data.

Transaction costs are a critical issue, especially for moving average strategies. These strategies hold positions continuously, leading to frequent position changes and higher sensitivity to transaction fees. A single long-to-short turnaround with the default cost of 0.1% incurs a total cost of 0.2% (entry + exit). This cost scales proportionally with the transaction fee level, reaching 1% at the highest cost of 0.5% analyzed in this research

This study used a 0.1% cost assumption for all calculations. This fee level is not tied to a specific exchange rate but reflects a commonly observed level offered by the largest cryptocurrency exchanges.

While some exchanges may offer lower fees under specific circumstances, the 0.1% assumption takes into account not only transaction fees but also potential slippage and changing bid-ask spread. This broader consideration aims to provide a realistic estimate of trading costs in

real-world scenarios.

To assess the impact of transaction costs on the strategy’s performance, this study analyzed results under seven different cost levels: 0.05%, 0.07%, 0.10%, 0.20%, 0.30%, 0.40%, 0.50%

This study investigated the impact of transaction costs on the performance of the Exponential Moving Average Crossover strategy. The strategy generated signals to enter long or short positions, and each position change incurred a transaction cost. Notably, switching from a long position to a short position requires closing the long position before opening the short, resulting in double value of the transaction cost for such a turnaround.

To assess cost sensitivity, the analysis applied varying transaction cost levels throughout the global training period to the original positions list generated by the strategy. While this approach might not perfectly replicate real-world scenarios where higher costs incentivize strategies with fewer trades, it provides a valuable starting point for understanding how transaction costs affect high-frequency strategy performance with all other factors held constant.

4.6 Algorithms descriptions

This section delves into the details of the algorithm employed in the study. The algorithm itself is presented in pseudocode format

4.6.1 WF parameters optimization algorithm with evaluation of results on the unseen period

Algorithm 1 outlines the steps involved in optimizing parameters for the walk-forward optimization process. To enhance clarity, this algorithm has been segmented into two distinct parts. The first part, presented here, focuses on the optimization of training and testing window sizes within the walk-forward framework. The second part, detailed in section The Walk-Forward procedure algorithm, describes the actual walk-forward procedure itself.

1. **Looping Through Parameter Combinations:** The algorithm iterates through all possible combinations of training and testing window sizes, which are the key parameters in the optimization process. This process is described in Algorithm 1.
2. **Walk-Forward Execution:** The walk-forward procedure is executed for each parameter combination. This step constitutes the core trading logic, involving the calculation of the EMA at each walk-forward step. A detailed description of the walk-forward algorithm can be found in Algorithm 2.

3. Sharpe Calculation: The performance of each walk-forward evaluation is measured using the Sharpe Ratio, a metric that considers both returns and risk.
4. Sharpe Ratio Grid Construction: Once all combinations are tested, the algorithm constructs a “grid” using the obtained Sharpe Ratios. Each column in this grid represents the training window size used, while each row represents the testing window size.
5. Smoothing the Sharpe Ratio Grid: This step involves refining the Sharpe Ratio grid by applying a smoothing technique. This might involve incorporating both the original Sharpe Ratio value and the values of its neighboring cells in the grid into the calculation.
6. Selecting Optimal Combinations: Finally, the two combinations of training and testing window size that lead to the highest Sharpe Ratios in the smoothed grid are chosen.
7. Applying Strategy to Unseen Data Period: These two chosen parameter combinations are then used to evaluate the walk-forward strategy on the Unseen Data Period for three specific cryptocurrencies: Bitcoin (BTC), Binance Coin (BNB) and Ethereum (ETH).

Algorithm 1: WF Parameters opitmization

```

1: procedure WFPARAMETERSOPTIMIZATION(DataSeries)
2:   Divide DataSeries into globalTrain, unseenTest
3:   Define trainCoin used in global train optimization
4:   Define testCoins used in unseenTest
5:   Define WFPairs(train, test) which consists of a combination of Walk-Forward train/test
   length period
6:   Define matrix WFGlobalTrainSharpe which will store Sharpe for each combination of
   train/test
7:   for each WFPair(train, test) in WFPairs do
8:     Call WFCalculateSharpe(globalTrain for trainCoin, WFPair(train, test)) proce-
     dure and store results in WFGlobalTrainSharpe
9:     From WFGlobalTrainSharpe get 2 MaxWFPairs(train, test) with highest Sharpe
10:    for each WFPair(train, test) in MaxWFPairs do
11:      Get unseenTest for testCoins
12:      Call WFCalculateSharpe(unseenTest, WFPair(train, test)) procedure and store re-
      sults in UnseenResults
13:    Report UnseenResults

```

4.6.2 The Walk-Forward procedure algorithm

The walk-forward procedure is executed for each combination of training/testing window sizes during the walk-forward parameter optimization process, which is detailed in section 1.

1. Data Segmentation: Involves segmenting the historical data series into walk-forward steps. Each walk-forward step has a length equal to the combined size of the training window and the testing window. The specific combination of training window size and testing window size is provided as a parameter when defining the walk-forward optimization process.

2. **Parameter Definition:** Define combinations of fast and slow EMA window sizes for the Exponential Moving Average (EMA) strategy. These parameters will be optimized for each walk-forward training period.
3. **Iterating through walk-forward steps:** The procedure iterates through each data walk-forward step containing both a training period and a testing period.
4. **Training Data Optimization:** Within each segment, calculate returns using the EMA strategy for various combinations of fast and slow EMA window sizes applied to the training data.
5. **Performance Evaluation:** For each combination of EMA periods, calculate the Sharpe Ratio based on the strategy's returns on the training data.
6. **Optimal Combination Selection:** After evaluating all EMA parameter combinations, choose the one that leads to the highest Sharpe Ratio as the optimal combination for that specific walk-forward training period.
7. **Testing with Optimal Parameters:** Using the chosen EMA parameters (fast and slow periods), calculate the strategy's returns on the testing data within the current walk-forward step.
8. **Test Result Accumulation:** Store the returns achieved in the testing period for further analysis.
9. **Segment-wise Repetition:** Repeat steps 3-8 for all steps in the walk-forward process.
10. **Overall Performance Calculation:** Finally, calculate the overall Sharpe Ratio for the entire walk-forward process, considering the combined returns from all testing periods across all walk-forward steps.

Algorithm 2: walk-forward (WF) Procedure

```

1: procedure WFCALCULATESHARPE(DataSet,train,test)
2:   define MaPairs(fast,slow)
3:   Divide DataSet into WFSegments with length train + test
4:   Initialize list WFReturns to keep overall returns of the whole WF
5:   for each segment(WFTrain, WFTest) in WFSegments do
6:     Define empty SharpeList for keeping the Sharpe ratio for each MA combination on
the testing period
7:     for each MaPair(fast,slow) in MaPairs do
8:       calculate EmaReturns using MaPair(fast,slow) on WFTrain
9:       calculate Sharpe using EmaReturns
10:      store Sharpe with MaPair(fast,slow) in the SharpeList
11:    End For
12:    Get BestMaPair(fast,slow) from SharpeList with highest Sharpe
13:    calculate EMATestReturn using BestMaPair(fast,slow) on WFTest
14:    calculate SharpeTest using EMATestReturns
15:    append EMATestReturns to WFReturns
16:  End For
17:  calculate WFTotalSharpe using WFReturns
18:  return WFTotalSharpe

```

4.6.3 Performance metrics

The following metrics are used across the study to measure profitability and risk.

Appropriate performance statistics presented in Table 2 were selected based on Ryś and Ślepaczuk 2019, which extensively describes the process of creating automated trading systems and the evaluation of strategies' performance.

Table 2: Performance statistics definitions.

Metric	Formula	Additional information
Log return	$R_t = \ln(\frac{p_t}{p_{t-1}})$	p_t the asset price at time t
Annualized Mean Return	$(\frac{1}{n} \sum_{t=1}^N R_t) * n_{year}$	n_{year} Number of log returns in the year N number of log returns in the strategy
Annualized Volatility	$\sqrt{\frac{1}{N-1} \sum_{t=1}^N (R_t - \bar{R})^2} * \sqrt{n_{year}}$	$\bar{R}$ average log return
Sharpe ratio	$\frac{\text{Annualized Mean Return} - r_{\text{risk free}}}{\text{Annualized Volatility}}$	$r_{\text{risk free}}$ Risk Free Rate in this study is assumed to be 0
Maximum Drawdown	$\max_{\tau \in [0, T]} \max_{t \in [\tau, T]} (p_\tau - p_t, 0)$	Worst cumulative loss from a local maximum p_t logarithm of asset price $\ln(P_t)$ at the time t and $t \in [0, T]$ P_t Asset price at time t
Information Ratio**	$\frac{\text{sign}(\text{Annualized Mean Return}) * \text{Annualized Mean Return}^2}{\text{Annualized Volatility} * \text{Maximum Drawdown}}$	
Sortino Ratio	$\frac{\bar{R} - r_{\text{benchmark}}}{\sqrt{\sigma_{\text{Negative log Returns}}}} * \sqrt{n_{year}}$	$r_{\text{benchmark}}$ Benchmark return in this study assumed to be 0 $\sigma_{\text{Negative log Returns}}$ Standard deviation of downside log returns, lower than the assumed level of $r_{\text{benchmark}}$

5 Results - Global Training Data Period

5.1 Choice of data frequency on results

This study employed an exhaustive search to identify the optimal walk-forward (WF) length for each timeframe tested. Essentially, for every timeframe (1, 5, 10, 15, 30, and 60 minutes), the walk-forward was optimized across all predefined training and testing period length combinations. For each interval, the walk-forward performance was estimated using the same combination of training and testing lengths for all window sizes (1, 2, 3, 5, 7, 10, 14, 21, 28). This resulted in 81 strategies per timeframe. Each combination yielded a Sharpe Ratio, reflecting the strategy's risk-adjusted performance. Table 3 summarizes the descriptive statistics of these Sharpe ratios for each timeframe. This provides a comprehensive overview of the strategy's performance across different sampling frequencies.

Key findings:

- Higher frequency timeframe (1-15) might be unprofitable due to 0.1% transaction costs, which were assumed
- Among the tested timeframes, 15- and 30-minute intervals contain some WF combinations with positive returns.
- Notably, the 60-minute timeframe also shows a positive mean Sharpe ratio
- No negative Sharpe at 60-minute frequency, even if the strategy is chosen randomly in 60 60-minute timeframe, it still exhibits a positive Sharpe Ratio

Table 3: The Descriptive Statistics of Sharpe Ratio for 81 walk-forward combinations per timeframe - Global Train Data Period

Frequency Minutes	Mean Sharpe	Max Sharpe	Min Sharpe	STD Sharpe	Q25% Sharpe	Q50% Sharpe	Q75% Sharpe
1	-12.7144	-11.4142	-16.6509	1.0943	-12.7549	-12.5585	-12.1700
5	-2.8416	-1.6011	-5.0528	0.8268	-3.0947	-2.6447	-2.3863
10	-1.4896	-0.4917	-2.9345	0.6626	-1.9249	-1.4718	-1.0263
15	-0.9783	0.1951	-2.7203	0.7177	-1.1952	-0.7957	-0.5273
30	-0.4954	0.7372	-2.1529	0.6440	-0.9490	-0.4528	-0.1158
60	0.7908	1.2524	0.1863	0.2256	0.6479	0.8124	0.9540

Note: The descriptive statistics of the Sharpe Ratio achieved by Exponential Moving Averages (EMA) strategies with walk-forward optimization executed on different time frequencies within the global training data, using varying training and testing window sizes. For each time frequency, the walk-forward optimization was performed multiple times with the same window combinations. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

5.2 The walk-forward optimization results

5.2.1 Walk-Forward Sharpe Ratios: Heatmap by Training/Testing Length (60-Minute)

Figure 2 presents a heatmap visualizing the Sharpe ratios achieved through a walk-forward optimization process applied to 60-minute Bitcoin price data within the global training dataset. The columns represent the lengths of the training periods, while the rows represent the lengths of the testing periods. Each cell in the heatmap displays the Sharpe ratio achieved using a specific combination of training and testing period lengths within the global training data.

The heatmap could be divided into four quarters:

First Quadrant (Bottom-Left):

- This quadrant corresponds to short walk-forward testing lengths (around 1 to 7 days) and short walk-forward training lengths (around 1 to 7 days).

Key insights:

- Generally low Sharpe ratios.
- Best training period: 5 days.
- Best testing period: 7 days.

- Surprisingly, combinations ≤ 3 days yield the lowest Sharpe ratios, suggesting frequent retraining might not always be beneficial.
- For 1-day training, testing periods of 5-7 days show the best results

Second Quadrant (Top-Left):

- Walk-forward training lengths (around 1 to 7 days) and longer walk-forward testing lengths (around 8 to 28 days).

Key insights

- The highest Sharpe ratios across all training lengths were observed for testing periods of 28 days.
- Best training periods: 5-7 days.
- Best performance: 7-day training and 28-day testing.

Third Quadrant (Bottom-Right):

- Walk-forward training lengths (around 10 to 28 days) and longer walk-forward testing lengths (around 1 to 7 days).

Key insights:

- Only 2 combinations with training periods of 10 and 14 days have Sharpe ratios exceeding 1.
- 1-day testing period consistently yields the lowest Sharpe ratios.

Fourth Quadrant (Top-Right):

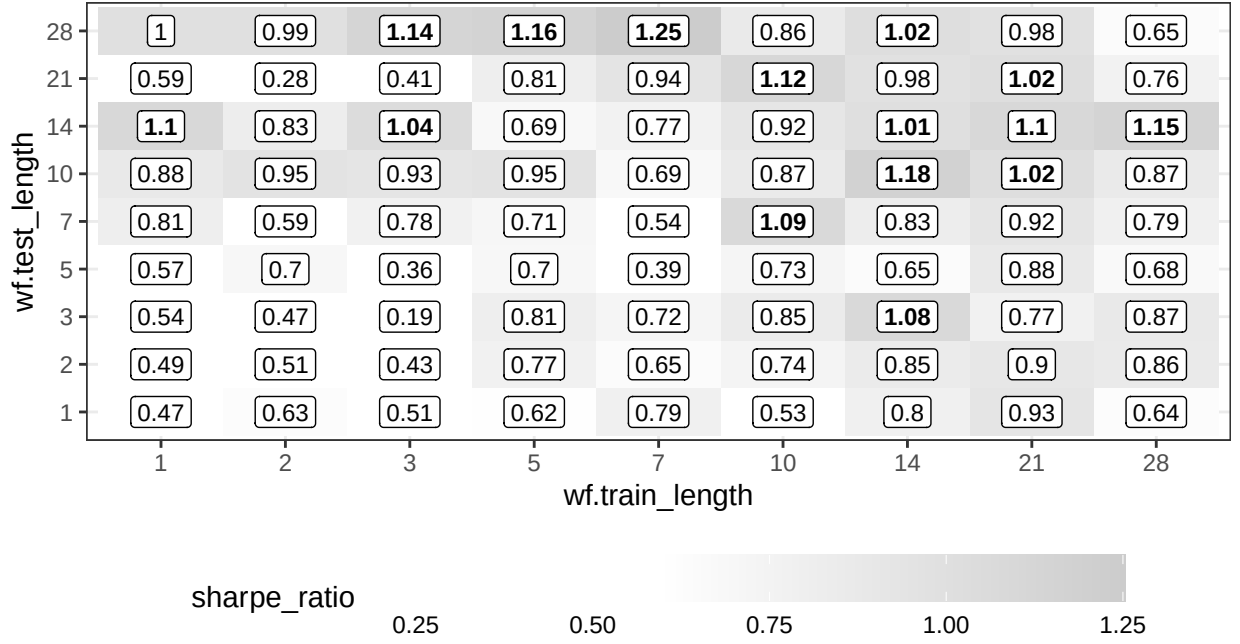
- Walk-forward training lengths (around 10 to 28 days) and longer walk-forward testing lengths (around 10 to 28 days)

Key insights:

- 8 combinations achieve Sharpe ratios above 1.
- Best performance: 14-day training and 10-day testing.

Overall: Quadrant 4 (long training, long testing) surprisingly exhibits the highest Sharpe ratios. This finding seems counterintuitive because less frequent retraining might prevent the strategy from adapting quickly to changing market conditions. Further investigation is necessary to understand this outcome.

Figure 2: Walk-Forward Sharpe Ratios: Heatmap by Training/Testing Length



Note: This graph analyzes the Sharpe Ratios for walk-forward optimization on 60-minute Bitcoin price data within the global training period, using Exponential Moving Averages (EMA). The horizontal axis represents the training window size, which is the amount of historical data used to train the EMA strategy. The vertical axis represents the testing window size, which is the amount of data used to evaluate the performance of the EMA strategy after training. The intersection of a specific training window size and testing window size on the chart shows the corresponding Sharpe Ratio achieved by the walk-forward optimization with EMA for those parameters. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%.

5.2.1.1 Walk-Forward Robust Sharpe Ratios: Heatmap (Smoothed Values) by Training/Testing Length (60-Minute)

Figure 3 presents a heatmap with a Robust Sharpe Ratio derived from the data in Figure 2. The smoothing formula proposed by Gómez and Ślepaczuk (2021) is described in section Choice of Smoothing Weights. This technique takes into account not only each value's original position but also the values of its neighboring cells, resulting in a clearer visualization of overall trends. Following the work of Gómez and Ślepaczuk (2021), the formula for the calculation of the Robust Sharpe Ratio for a given cell in the heatmap is presented here (4):

$$\text{Robust Sharpe Ratio} = \frac{1}{2} * \text{Sharpe Ratio}_{\text{Original}} + \frac{1}{2} * \sum_{n=1}^n \frac{1}{n} \text{Sharpe Ratio}_{\text{Neighbour}} \quad (4)$$

where :

n – the index of the neighbor.

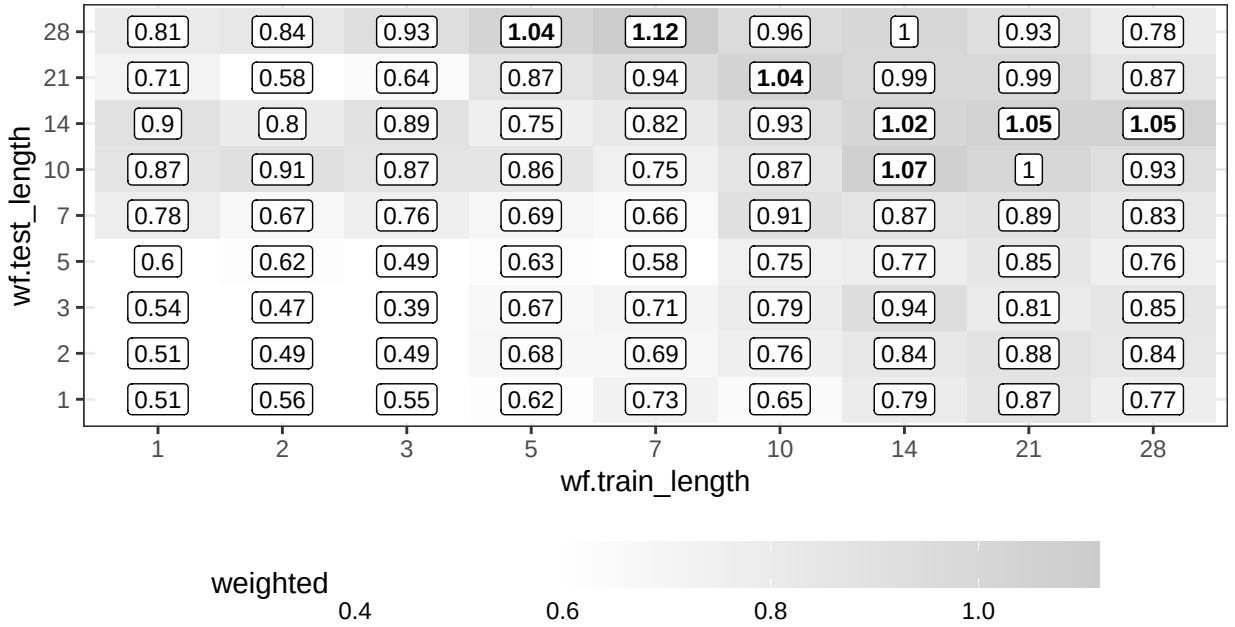
$\text{Sharpe Ratio}_{\text{Original}}$ – Sharpe Ratio for given combination of training/testing period

$\sum_{n=1}^n \frac{1}{n} \text{Sharpe Ratio}_{\text{Neighbour}}$ – The sum of all neighbors

Key insights:

- The highest Sharpe Ratios are observed in the area with longer training periods (14-21-28 days), specifically for 14-days and 10-days testing periods
- Based on this Robust Sharpe Ratio heatmap, two combinations were selected for further testing on Unseen Data Period:
 - training 7 days testing 28 days
 - training 14 days testing 10 days
- It's important to note that this selection was somewhat subjective, as it assumed the best training period performance translates to the best performance on the Unseen Data Period. This is not guaranteed.
- Alternative selection criteria could be explored, such as choosing combinations with the most stable results across different training and testing lengths.

Figure 3: Walk-Forward Robust Sharpe Ratios: Heatmap (Smoothed Values) by Training/Testing Length



Note: This graph analyzes the smoothed heatmap of Robust Sharpe Ratios for walk-forward optimization on 60-minute Bitcoin price data within the global training period, using Exponential Moving Averages (EMA). The horizontal axis represents the training window size, which is the amount of historical data used to train the EMA strategy. The vertical axis represents the testing window size, which is the amount of data used to evaluate the performance of the EMA strategy after training. The intersection of a specific training window size and testing window size on the chart shows the corresponding Sharpe Ratio achieved by the walk-forward optimization with EMA for those parameters. Robust Sharpe Ratio values were calculated using a weighted average. In this approach, the original value retained half the weight, while the remaining weight was distributed among its neighboring values. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%.

5.2.1.2 Selected Combinations

This study selected parameters for the Unseen Data Period evaluation based on their performance in the global training period. Specifically, the two parameter sets that achieved the highest smoothed Sharpe Ratios (presented in Figure 3) were chosen.

However, it is important to acknowledge potential alternative choices. An astute reader might identify combinations like 21/14 on the smoothed Sharpe grid, which appears visually more promising due to its location within the central “greyish area” of the heatmap, indicating a

higher smoothed Sharpe ratio compared to the chosen 14/10.

Furthermore, the choice of 7/28 days raises another potential concern. This selection resides on the edge of the search window, with estimates based on only 5 neighboring cells instead of the usual 8. This introduces a higher degree of uncertainty compared to centrally located choices.

Despite these considerations, the study emphasizes that the parameter selection process aimed for simplicity, prioritizing the two best-performing parameter sets based on the initial smoothing pass.

The following combinations achieved the highest Sharpe Ratios in the Robust Sharpe Ratio Heatmap (presented in Figure 3) and were chosen for further evaluation:

- Training 7 days, Testing 28 days
- Training 14 days, Testing 10 days

5.3 Performance Metrics - Global Training Data Period - Walk-Forward Top Combinations

Table 4 summarizes performance metrics for trading strategies identified through a walk-forward optimization process, where the length of the training and testing periods of a walk-forward was optimized. These strategies, which use Exponential Moving Averages (EMA) achieved the highest Sharpe Ratio for Bitcoin within the global training period. The process of selecting of optimal walk-forward parameters was described in the Selected Combinations section.

Key findings:

- Annualized Mean Return: Both strategies achieved similar returns, with a difference of only 5% compared to the overall 94% return. This difference was considered negligible.
- Annualized Volatility: Both strategies exhibited identical volatility.
- Sharpe Ratio: The 7/28 combination (7 days training, 28 days testing) had a higher Sharpe ratio due to its slightly higher annualized return.
- Maximum Drawdown: The 7/28 combination significantly outperformed the other with a much smaller maximum drawdown, indicating better downside risk protection.
- Information Ratio**: The 7/28 combination (training 7 days, testing 28 days) achieved a higher Information Ratio compared to alternatives, driven by its superior performance in both return and drawdown.
- Sortino Ratio: While marginally higher for the 7/28 strategy, this was likely related to its better performance in terms of drawdown.

Table 4: Trading Metrics: Top Sharpe Ratio Strategies (Walk-Forward, Global Training)

Description	Annualized Mean Return	Annualized Volatility	Sharpe Ratio	Information Ratio**	Max Drawdown	Sortino Ratio
BTC Train 7 Test 28	0.9483	0.7572	1.252	4.622	0.3516	1.799
BTC Train 14 Test 10	0.8918	0.7571	1.178	2.902	0.4513	1.678

Note: This table presents trading metrics for strategies identified through a walk-forward optimization process, where the length of the training and testing periods of a walk-forward was optimized. These strategies use Exponential Moving Averages (EMA), achieved the highest Sharpe Ratio for Bitcoin within the global training period. The global training data period covers data from February 8, 2018, to September 1, 2019. Each row showcases the performance metrics when using varying training and testing window sizes for a walk-forward optimization. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

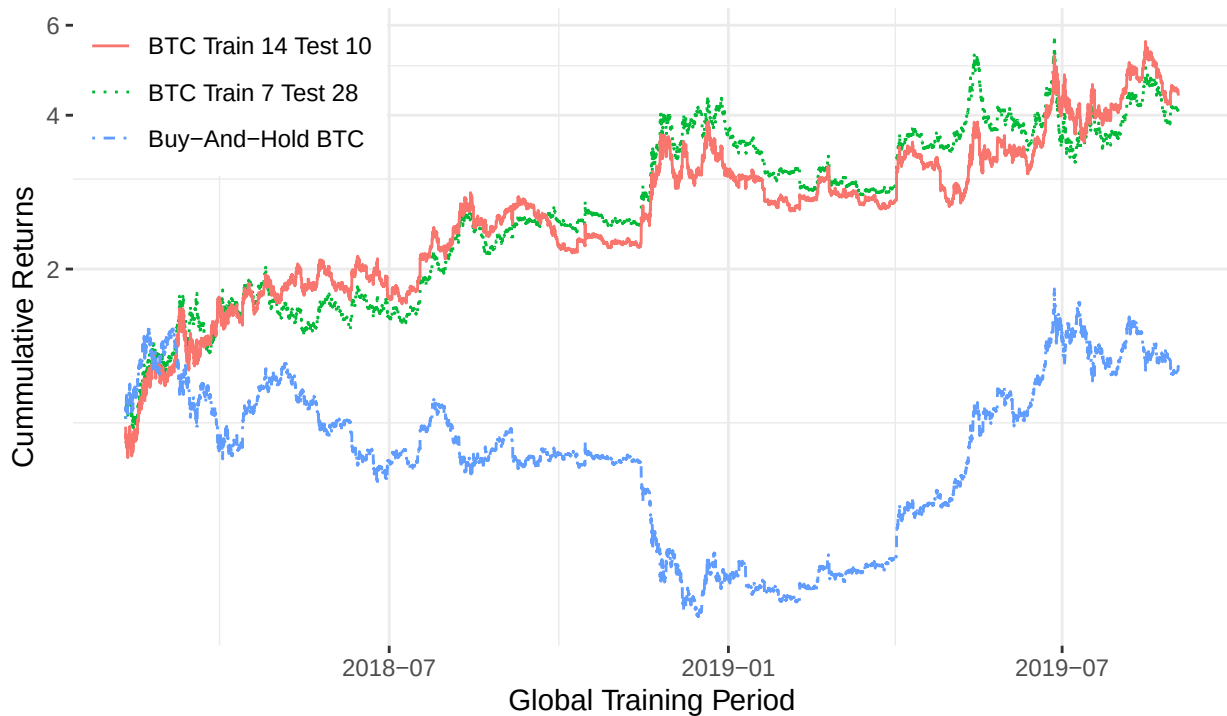
5.4 Equity Curves - Global Training Data Period - Walk-Forward Top Combinations

Figure 4 compares the cumulative returns (equity curves) of a Buy-and-Hold strategy with trading strategies identified through a walk-forward optimization process, where the length of the training and testing periods of a walk-forward was optimized. These strategies, using Exponential Moving Averages (EMA), achieved the highest Sharpe Ratio for Bitcoin within the global training period. Buy-and-Hold strategy assumes the purchase of a given asset at the beginning and selling it at the end of the period. The process of selecting of optimal walk-forward parameters was described in the Selected Combinations section.

Key findings:

- Both EMA strategies exhibited similar overall performance, outperforming the Buy-and-Hold strategy.
- The EMA strategies tended to be more profitable when Bitcoin prices declined, suggesting they capitalized on short-term trends.
- Conversely, when Bitcoin prices recovered (e.g., after the Bitcoin decline in early 2019), the EMA strategies underperformed the Buy-and-Hold.
- Most of the gains for the EMA strategies occurred in the first half of the period, while the second half saw them lagging behind the recovering Buy-and-Hold.

Figure 4: Equity Curves: Top Sharpe Ratio Strategies (Walk-Forward, Global Training Data Period)



Note: Cumulative Returns (Equity curves) for strategies identified through a walk-forward optimization process, where the length of the training and testing periods of a walk-forward was optimized. These strategies, used Exponential Moving Averages (EMA), achieved the highest Sharpe Ratio for Bitcoin within the global training period. Buy-and-hold strategy assumed the purchase of given asset at the beginning and selling it at the end of period. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

5.5 Statistical significance - Bootstraps

To assess the effectiveness of the chosen parameter combinations for a walk-forward optimization and evaluate their statistical significance, two bootstrap procedure tests were performed. Details of the procedures are described in the Statistical significance section.

5.5.1 Results - Bootstrap Evaluation of Subsampled EMA Combinations for Walk-Forward Optimization

Both strategy combinations selected in the Selected Combinations section were evaluated for statistical significance using the method detailed in the Bootstrap Evaluation of Subsampled EMA Combinations for Walk-Forward Optimization section. This method differs from the standard approach by using random EMA parameters drawn from a predefined set described in the section Exponential Moving Average Crossover, whereas the standard approach selects the best EMA parameters identified during the training period for each walk-forward step. This allows assessing the performance of the selection method compared to random selection. A bootstrap procedure was performed during the global training period.

Table 5: Results - Bootstrap Evaluation of Subsampled EMA Combinations for Walk-Forward Optimization

Description	Original Sharpe Ratio	Total No. of bootstrap iterations	No. Bootstrap Iterations with Higher Sharpe Ratio	Significance%
BTC Train 7 Test 28	1.252	1000	80	8.0
BTC Train 14 Test 10	1.178	1000	137	13.7

Note: Original Sharpe Ratio - Sharpe Ratios of the Pre-Selected Best Strategies from the Global Training Period, Total No. of bootstrap iterations - Number of times the bootstrap procedure was repeated, No. Bootstrap Iterations with Higher Sharpe Ratio - Number of bootstrap iterations that achieved a Sharpe Ratio exceeding the original strategy's Sharpe Ratio. For clarification, "original" in this context refers to the strategies that were pre-selected during the global training period, as described in section Selected Combinations. This method differs from the standard approach by using random EMA parameters drawn from a predefined set described in the section [Exponential Moving Average Crossover], whereas the standard approach selects the best EMA parameters identified during the training period for each walk-forward step. This allows assessing the performance of the selection method compared to random selection. A bootstrap procedure was performed on the global training period. The cost assumption remains the same as in previous calculations. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

Table 6 presents the descriptive statistics for Sharpe Ratio achieved during bootstrap iterations.

Key findings:

- The average Sharpe Ratios across both bootstrap strategies are very close
- The maximum Sharpe Ratio is 40-50% higher than the original strategies Sharpe Ratios
- The original strategies' Sharpe Ratios fall within 1 standard deviation of the average Sharpe Ratio across bootstrap iterations. For clarification, "original" in this context refers to the strategies that were pre-selected during the global training period, as described in section Selected Combinations.

Table 6: The Descriptive Statistics of Sharpe Ratios - Bootstrap Evaluation of Subsampled EMA Combinations for Walk-Forward Optimization

Description	Mean	SD	Median	Min	Max	Range	Skew	Kurtosis
BTC Train 7 Test 28	0.7847	0.3259	0.7888	-0.4584	1.777	2.236	-0.0856	0.1486
BTC Train 14 Test 10	0.7901	0.3448	0.7948	-0.3549	1.716	2.071	-0.1403	-0.1138

Note: This method differs from the standard approach by using random EMA parameters drawn from a predefined set described in the section [Exponential Moving Average Crossover], whereas the standard approach selects the best EMA parameters identified during the training period for each walk-forward step. This allows assessing the performance of the selection method compared to random selection. A bootstrap procedure was performed on the global training period. The cost assumption remains the same as in previous calculations. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

5.5.1.1 Results—Bootstrap Evaluation of Shuffled Transaction blocks Both strategy combinations selected in the Selected Combinations section were evaluated for statistical significance using the method detailed in the [Bootstrap Evaluation of Random EMA Combinations in Walk- Forward Optimization] section. This method of bootstrap shuffles blocks of original transactional positions generated by the walk-forward within the global training period.

Key findings:

- Both combinations were statistically significant
- 5% statistical significance was reached by both sets of strategies.

Table 7: Results - Bootstrap Evaluation of Shuffled Transaction Blocks

Description	Original Sharpe Ratio	Total No. of bootstrap iterations	No. Bootstrap Iterations with Higher Sharpe Ratio	Significance%
BTC Train 7 Test 28	1.252	1000	35	3.5
BTC Train 14 Test 10	1.178	1000	44	4.4

Note: Original Sharpe Ratio - Sharpe Ratios of the Pre-Selected Best Strategies from the Global Training Period, Total No. of bootstrap iterations - Number of times the bootstrap procedure was repeated, No. Bootstrap Iterations with Higher Sharpe Ratio - Number of bootstrap iterations that achieved a Sharpe Ratio exceeding the original strategy's Sharpe Ratio. For clarification, "original" in this context refers to the strategies that were pre-selected during the global training period, as described in section Selected Combinations. Bootstrap Evaluation of Shuffled Transaction Blocks shuffles blocks of original transactional positions generated by the walk-forward within the global training period. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

Table 8 presents the descriptive statistics of the Sharpe Ratios for random strategies generated through bootstrapping.

Key findings :

- The maximum Sharpe Ratio achieved here is significantly higher than that reached by the original strategy. This suggests there could be a much better method for selecting the EMA for the walk-forward testing period than simply selecting the combination that performs best in the walk-forward training period.
- The bootstrap based on shuffled positions from the 7/28 strategy has a smaller range and maximum value compared to the 14/10 strategy.
- The average Sharpe Ratio across bootstrap iterations falls below zero for both strategies.
- Standard Deviation is much higher than in the other bootstrap method
- The original strategies' Sharpe Ratios fall within 1 standard deviation of the average Sharpe Ratio across bootstrap iterations. For clarification, "original" in this context refers to the strategies that were pre-selected during the global training period, as described in section Selected Combinations.

Table 8: Descriptive Statistics of Sharpe Ratios - Bootstrap Evaluation of Shuffled Transaction blocks

Description	Mean	SD	Median	Min	Max	Range	Skew	Kurtosis
BTC Train 7 Test 28	-0.2373	0.8083	-0.2391	-2.789	2.273	5.062	-0.0440	-0.0346
BTC Train 14 Test 10	-0.1737	0.8012	-0.1955	-3.098	2.721	5.819	-0.0435	0.1718

Note: Bootstrap Evaluation of Shuffled Transaction blocks shuffles blocks of original transactional positions generated by the walk-forward within the global training period. The global training data period covers data from February 8, 2018, to September 1, 2019. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

5.6 Cost Sensitivity

Recognizing the significant influence of cost on strategy outcomes, a cost sensitivity analysis investigated how the result characteristics change with varying cost levels, isolating cost as the sole variable.

5.6.1 Strategy Metrics - Cost sensitivity - Global Training Data Period

Key findings:

- Profitability: Increasing transaction cost per trade from 0.1% to 0.2% leads to a decrease in annualized profit, while volatility remains largely unchanged.
- Sensitivity to Cost Increases: Further increases of 0.1% per transaction (reaching 0.3%) result in an average 0.3% drop in Annualized Mean Return.
- Drawdown: Maximum drawdown is affected heavily by rising transaction costs.
- Information Ratio (IR): IR decreases non-linearly with rising cost.
- Profitability Threshold: The strategy becomes unprofitable at a cost level of around 0.4% per transaction.

Table 9: Cost sensitivity

Cost level	Annualized Mean Return	Annualized Volatility	Sharpe Ratio	Information Ratio**	Max Drawdown	Sortino Ratio
0.05%	1.1068	0.7569	1.4623	6.6457	0.3408	2.1035
0.07%	1.0419	0.7570	1.3763	5.7491	0.3442	1.9788
0.10%	0.9483	0.7572	1.2524	4.6218	0.3516	1.7990
0.20%	0.6197	0.7584	0.8171	1.5867	0.4240	1.1699
0.30%	0.2950	0.7604	0.3879	0.2902	0.4979	0.5531
0.40%	-0.0297	0.7633	-0.0390	-0.0024	0.6448	-0.0553
0.50%	-0.3545	0.7670	-0.4622	-0.2193	0.7788	-0.6519

Note: Strategy metrics for Exponential Moving Average strategy optimized with walk-forward process, with parameters of 7 days for training and 28 days for testing, applied in the global training period, assuming different transactional costs.

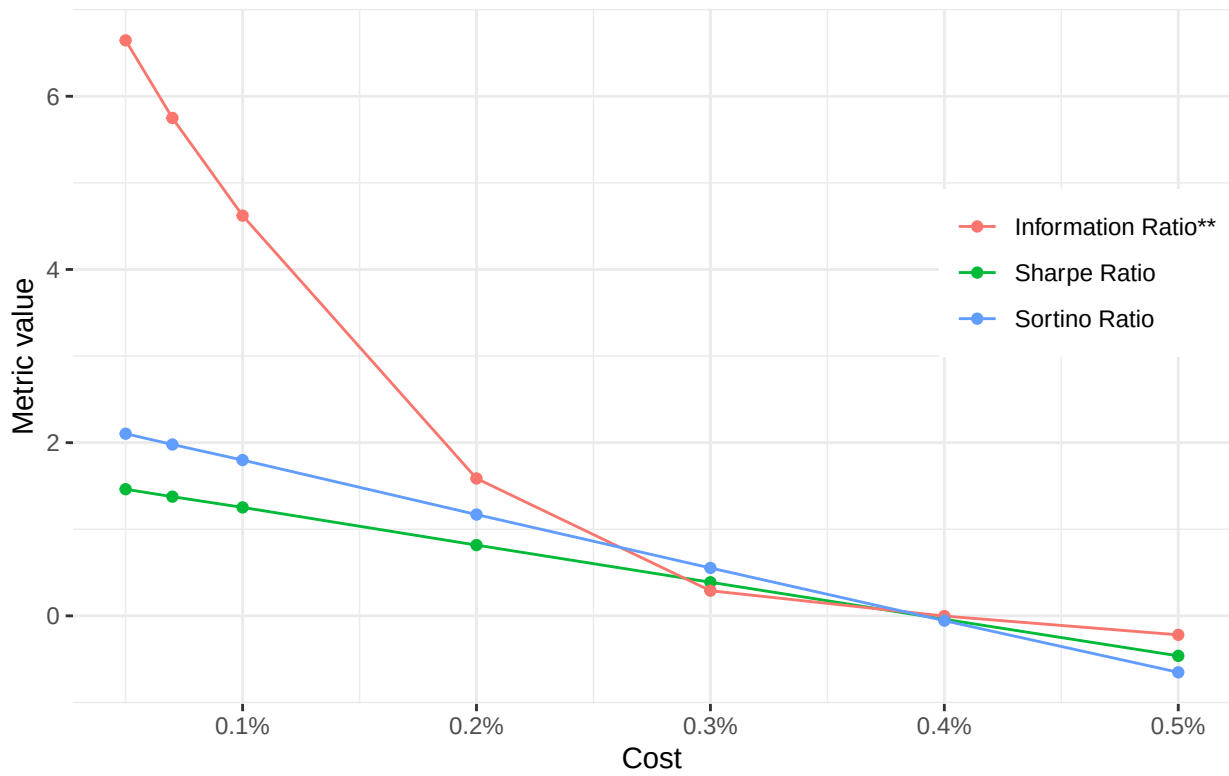
Strategy Metrics Visualization - Cost sensitivity - Global Training Data Period

To visualize the impact of transaction costs on the strategy's performance, metrics are presented graphically. The horizontal axis represents the level of transaction cost, while the vertical axis displays unitless metric values.

Key findings:

- Information Ratio** appears to be the metric most significantly affected by cost changes.
- The decrease in Information Ratio** is non-linear, with a steeper decline observed at the beginning of the cost range compared to the end.
- Other metrics exhibit a nearly linear decrease, with some fluctuations at the start of the cost range.
- Based on the observed trends, the breakeven point, where the strategy's performance breakeven transaction costs, can be estimated to be around 0.36% cost per transaction.

Figure 5: Strategy metrics for different level of transactional cost



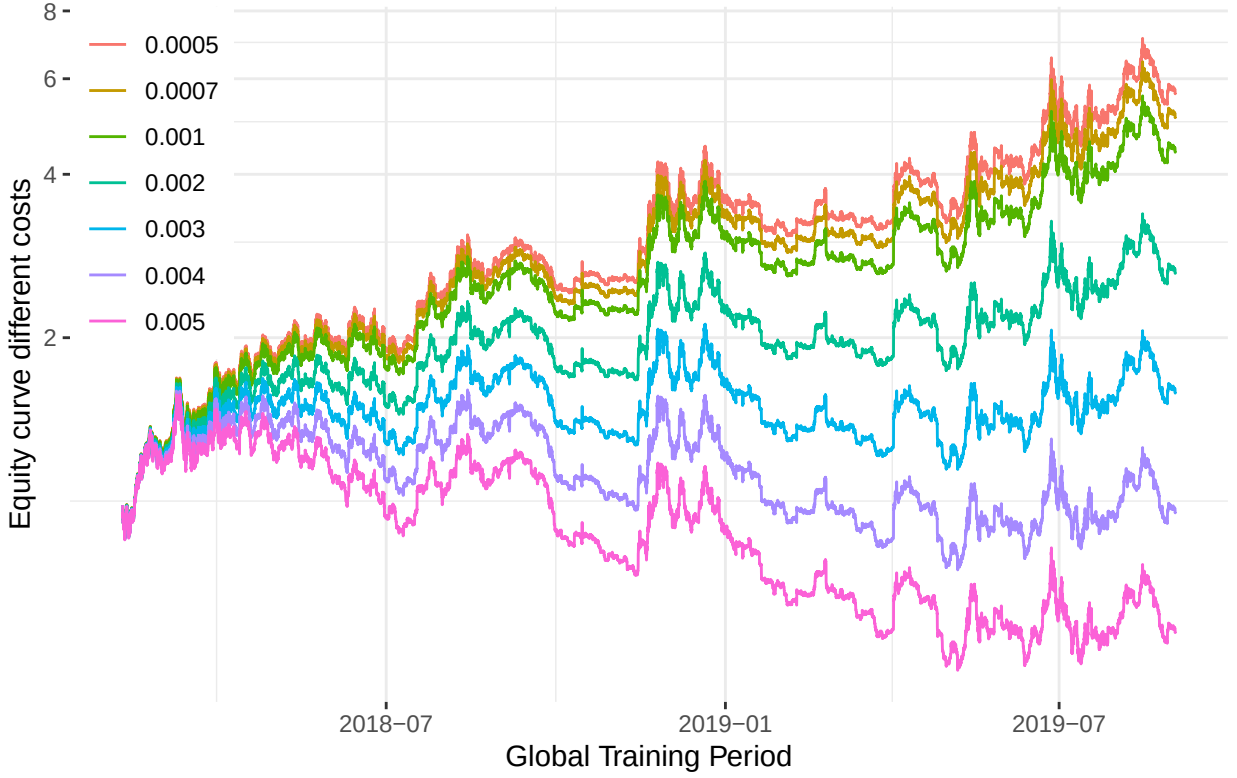
Note: Strategy metrics for Exponential Moving Average strategy optimized with walk-forward process, with parameters of 7 days for training and 28 days for testing, applied in the global training period, assuming different transactional cost.

Equity Curves - Cost sensitivity - Global Training Data Period

Key findings:

- Increasing cost changes the direction of the original equity curve to downward sloping
- Equity curves look the same, and their volatility did not increase just change direction
- The first equity curve, which is unprofitable, is for a cost of 0.4%, the breakeven point is somewhere between 0.3% to 0.4%

Figure 6: Equity curves for different level of transactional cost



Note: Equity curves for Exponential Moving Average strategy optimized with walk-forward process, with parameters of 7 days for training and 28 days for testing, applied in the global training period, assuming different transactional cost.

6 Results - Unseen Data Period

Following the optimization within the global training period, the two walk-forward parameter combinations yielding the highest Sharpe ratios were evaluated on unseen data. To further validate the method's robustness, the strategy was then applied to two other cryptocurrencies, Ethereum (ETH) and Binance Coin (BNB), which were not included in the global training period optimization. The selection of these additional assets was driven by their market capitalization. To mitigate the potential for cherry-picking bias, the strategy was evaluated on the unseen data period only once. This approach ensured that the results and parameters were not subjected to any manipulation or modification after the initial evaluation.

6.1 Results for single cryptocurrencies on unseen data

Table 10 shows metrics of trading strategies executed on the unseen data period. The walk-forward parameters of strategies were derived from the global training period. Combinations of parameters that achieved the highest Sharpe ratio were selected. Table 10 also includes a benchmark, which is Buy-and-Hold strategy.

The best walk-forward window size selected from the Global Training Data Period:

- Train 7 days test 28 days (WF 7/28)
- Train 14 days test 10 days (WF 14/10)

Key findings:

- Buy-and-Hold - Achieved the highest Sharpe Ratio for all cryptocurrencies, outperforming the walk-forward strategies for the given cryptocurrency Buy-and-Hold for ETH and BNB had higher Sharpe Ratios than BTC.
- Selected strategy performance - WF 14/10 outperformed the WF 7/28 strategy across all cryptocurrencies, with an average improvement of 10% in Sharpe Ratio for ETH and BNB, and a 40% higher Sharpe Ratio for BTC. Notably, both ETH and BNB were not included in the optimization process in the global training period, suggesting potential generalizability of the strategy.
- Comparison to Buy-and-Hold - No strategy surpassed the respective asset's Buy-and-Hold performance in terms of Sharpe ratio. For BTC, the WF 14/10 strategy came closest, with Buy-and-Hold outperforming by less than 2%. In contrast, ETH and BNB experienced a more significant performance gap, with Buy-and-Hold outperforming the WF 14/10 strategy by 20% and 15%, respectively. The WF 14/10 strategy outperformed Buy-and-Hold on Information Ratio** for ETH and by around 30% for BTC. For BTC only, the WF 14/10 strategy surpassed the performance of Sortino Buy-and-Hold.
- Performance by risk-adjusted return measure - (Sharpe Ratio, Sortino Ratio, Information Ratio**) Buy-and-Hold strategies for ETH and BNB achieved the highest Sharpe Ratios, followed by the WF 14/10 strategy in third place. BTC WF 14/10 strategy outperformed Buy-and-Hold in terms of the Sortino ratio, indicating better risk-adjusted returns. ETH and BNB - WF 14/10 strategy underperformed Buy-and-Hold by a smaller margin compared to the Sharpe ratio metric, with Buy-and-Hold outperforming by only 13% and 7% for ETH and BNB, respectively. In terms of Information Ratio**, Buy-and-Hold significantly outperformed the WF 14/10 by 35%, while the WF 14/10 reached the highest Information Ratio** among all strategies.

The overall observations:

- Buy-and-Hold remains a strong benchmark for all three cryptocurrencies, particularly in terms of Sharpe ratio.
- The WF 14/10 strategy shows promising results across different asset classes, exceeding other tested strategies and exhibiting potential generalizability.
- The choice of risk-adjusted return measure (Sharpe vs. Sortino) can influence the interpretation of performance, particularly when comparing strategies with different risk profiles.

Table 10: Performance Metrics - Unseen Data Period - Walk-Forward Top Combinations from Global Training Data Period

Description	Annualized Mean Return	Annualized Volatility	Sharpe Ratio	Information Ratio**	Max Drawdown	Sortino Ratio
BTC Buy and Hold	0.9267	<u>0.8214</u>	1.1281	2.0485	0.6228	1.554
BTC Train 7 Test 28	0.6828	0.8219	0.8307	1.3191	0.5849	1.2198
BTC Train 14 Test 10	0.9091	0.8216	1.1064	2.6275	<u>0.5243</u>	1.6249
ETH Buy and Hold	1.5771	1.0264	<u>1.5365</u>	4.0574	0.6937	<u>2.1089</u>
ETH Train 7 Test 28	1.2408	1.0268	1.2085	2.0651	0.7696	1.7669
ETH Train 14 Test 10	1.3727	1.0266	1.3371	<u>4.2635</u>	0.6177	1.9519
BNB Buy and Hold	<u>1.7142</u>	1.1705	1.4644	3.5801	0.7483	2.0211
BNB Train 7 Test 28	1.2879	1.1712	1.0997	1.8793	0.7798	1.6298
BNB Train 14 Test 10	1.403	1.171	1.1982	2.6565	0.7275	1.785

Note: Performance metrics for the combination of strategies with the highest Sharpe Ratio identified on the global training period and applied to the unseen period. The global training period was used in the case of BTC to find optimal walk-forward parameters and train/test day lengths of 7/28 and 14/10. Strategy with these combinations of parameters executed on the unseen data period using BTC and the cryptocurrencies not used before i.e., ETH and BNB. The bolded values indicate the best metric within three strategies for each cryptocurrency separately, while the underlined values indicate the best metrics globally for all nine strategies. The unseen data period covers data from November 7, 2019, to August 22, 2021. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

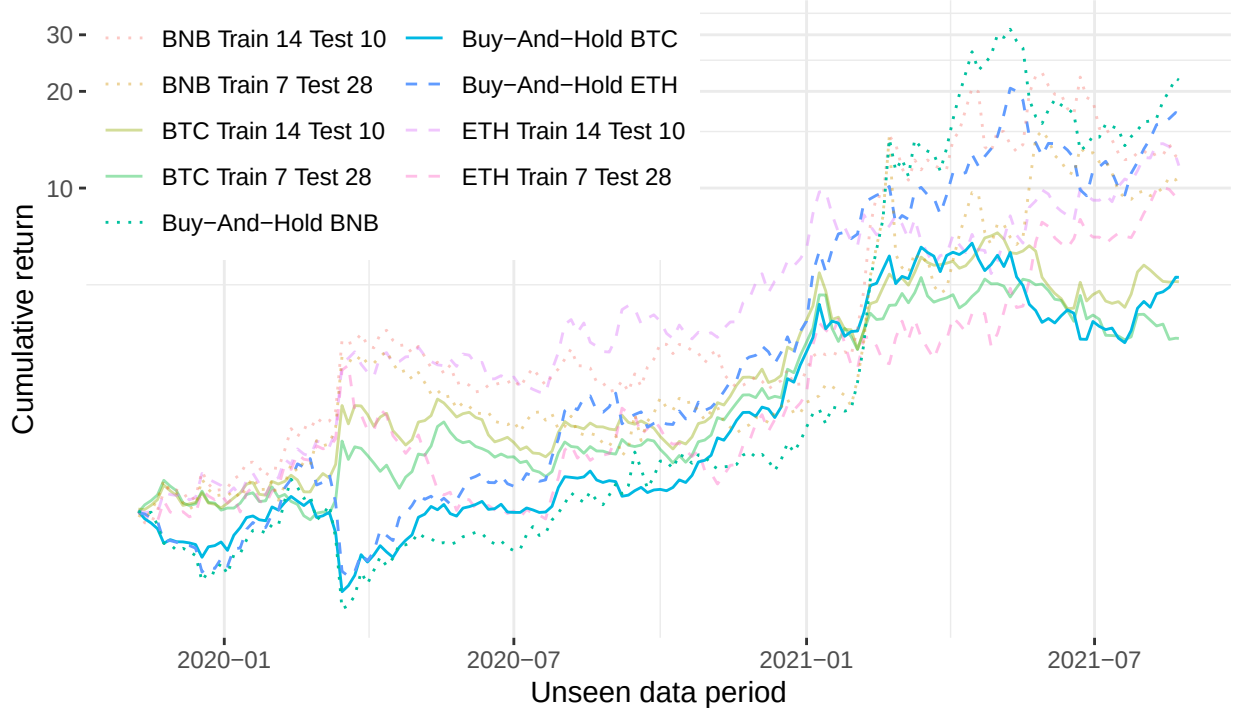
Equity Curves - Unseen Data Period - Walk-Forward Top Combinations from the Global Training Data Period

Key findings:

- BNB Buy-and-Hold finished with the highest cumulative gains of all strategies, underperforming in 2020 and overperforming in 2021.
- Both variations of strategy performed better than Buy-and-Hold in 2020; BNB WF 14/10 was unbeatable in 2020.
- The situation changed in 2021 when Buy-and-Hold take the lead and ETH, together with BNB, recovered all it loses.
- BTC strategies behaved very similarly to Buy-and-Hold. Buy-and-Hold underperformed throughout the entire period, only delivering some gains at the end of the unseen period. This resulted in a walk-forward strategies slightly underperforming Buy-and-Hold in terms of Sharpe Ratio
- All variations of strategies provided good protection against losses in 2020; when Buy-and-Hold lost significantly in March 2020, strategies observed gains.
- Problematic periods when the strategy started to lose steadily were the periods when Buy-and-Hold recovered and flat periods with low volatility.
- Interestingly, walk-forward strategies dominated throughout 2020, outperforming their Buy-and-Hold counterparts across all cryptocurrencies. If the study had concluded at the end of 2020, walk-forward strategies would likely have been the clear winner in every category. However, 2021 saw a shift, with walk-forward strategies underperforming against Buy-and-Hold strategies. The reason for this reversal remains unclear. However, it's

possible that walk-forward strategies based on parameters that were optimal in 2018 and 2019 are no longer providing an edge. Therefore, walk-forward strategies may benefit from retraining and recalibration to identify new optimal windows in changing market conditions.

Figure 7: Equity Curves - Unseen Data Period - Walk-Forward Top Combinations from Global Training Data Period



Note: Equity curves for the combination of strategies with the highest Sharpe Ratio identified on the global training period and applied to the unseen period. The global training period was used in the case of BTC to find optimal walk-forward parameters train/test days lengths 7/28 and 14/10. Strategy with these combinations of parameters executed on the unseen data period using BTC and the cryptocurrencies not used before i.e. ETH and BNB. The unseen data period covers data from November 7, 2019, to August 22, 2021. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

6.2 Results for portfolio of cryptocurrencies on unseen data

Three portfolios were tested on the Unseen Data Period to evaluate their performance:

- Buy-and-Hold: Benchmark portfolio holding Bitcoin (BTC), Ethereum (ETH), Binance Coin (BNB) no rebalancing during the whole period.
- Portfolio 1 (the walk-forward training 7 days/testing 28 days - WF 7/28): Strategy using an Exponential Moving Averages and the walk-forward with a 7-day training period and 28-day testing period, using 3 different cryptocurrencies: Bitcoin (BTC), Ethereum (ETH), Binance Coin (BNB). Equal weights were assigned initially, with no rebalancing during the whole period.
- Portfolio 2 (the walk-forward training 14 days/testing 10 days- WF 14/10): Similar strategy to Portfolio 1, but with a 14-day training period and 10-day testing period, applied to Bitcoin (BTC), Ethereum (ETH), and Binance Coin (BNB). Equal weights were assigned initially, with no rebalancing during the whole period.
- All portfolios combined: Buy-and-Hold, Portfolio 1, Portfolio 2 combined to assess the

potential performance benefits of combining active and passive strategies.

Performance Metrics - Portfolio - Unseen Data Period

Key finding:

- Sharpe Ratio: All portfolios combined reached the highest Sharpe Ratio, beating the second Buy-and-Hold portfolio by 20%. Combining all portfolios decreased volatility significantly.
- Information Ratio (IR): The combined portfolio strategy achieved the highest performance once again, surpassing second-placed WF 14/10 by a significant margin and doubling its value.
- Sortino Ratio: All portfolios combined scored the highest Sortino Ratio, outperforming the second WF 14/10, and the Buy-and-Hold portfolio by more than 30%.
- Maximum Drawdown: Combining all portfolios also resulted in the smallest maximum drawdown. Maximum drawdown decreased from nearly 70% for Buy-and-Hold to around 45%. This is particularly outstanding considering that walk-forward strategies experienced an average drawdown of around 60%.
- Combining all the portfolios significantly improved each performance metric. This resulted in decreased volatility and maximum drawdown and unleashed the power of diversification.

Table 11: Performance Metrics - Unseen Data Period - Walk-Forward Top Combinations from Global Training Data Period - Portfolio

Description	Annualized Mean Return	Annualized Volatility	Sharpe Ratio	Information Ratio**	Max Drawdown	Sortino Ratio
Portfolio 3 currencies Train 7 Test 28	1.129	0.889	1.27	3.047	0.619	1.874
Portfolio 3 currencies Train 14 Test 10	1.269	0.907	1.4	4.274	0.569	2.067
Portfolio 3 currencies Buy and Hold	1.495	0.97	1.542	3.87	0.683	2.075
All Portfolios Combined	1.318	0.686	1.921	9.369	0.438	2.757

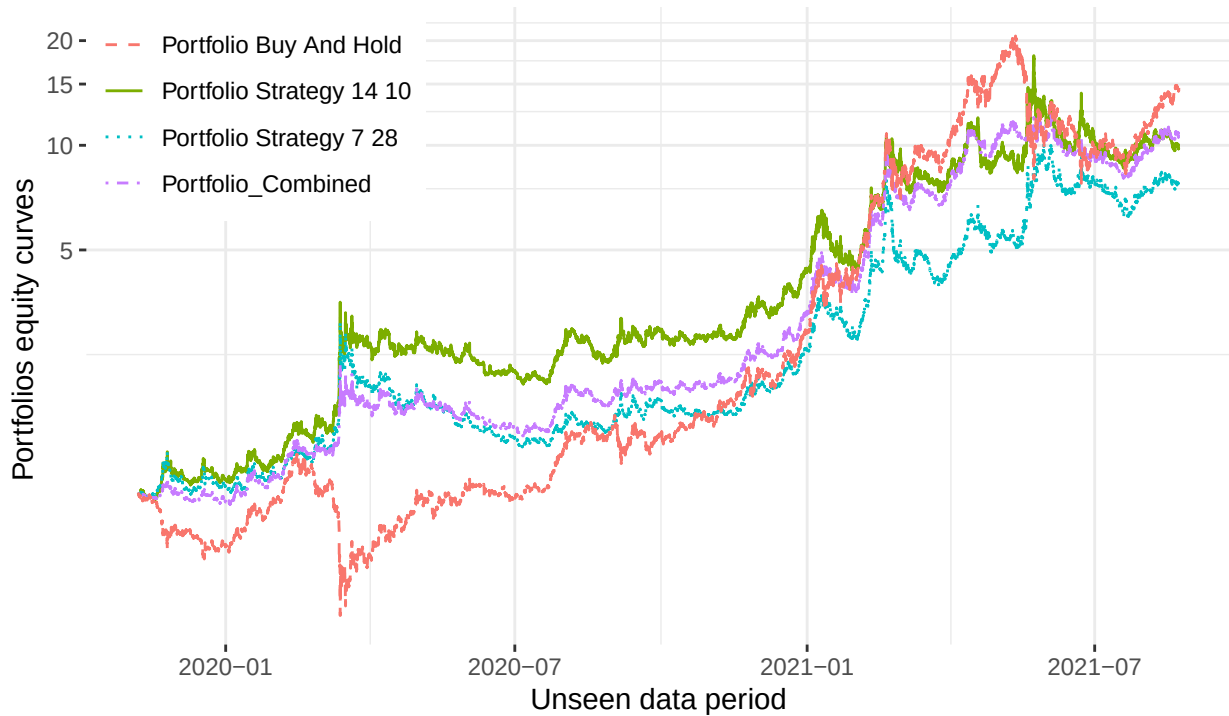
Note: Trading metrics for portfolios with equal weights at inception: 1. Buy-and-Hold portfolio - BTC, ETH, BNB equally weighted. 2. Portfolio - walk-forward strategy with training for 7 days and testing for 28 days - strategy applied to BTC, ETH, and BNB equally weighted. 3. Portfolio - walk-forward strategy with training for 14 days and testing for 10 days - applied to BTC, ETH, and BNB equally weighted. No rebalancing during the whole period. The unseen data period covers data from November 7, 2019, to August 22, 2021. The bolded values indicate the best metric within three strategies. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

Equity curves - Portfolio - Unseen Data Period

Key Findings:

- Profitability: Most profits for the MA/WF strategies occurred during periods of high volatility, coinciding with significant losses of the Buy-and-Hold portfolio.
- Recovery Phases: During Buy-and-Hold portfolio recovery periods, both WF portfolios lost momentum or declined.
- Upward Trends: In stable upward markets, the WF portfolios slightly underperformed the Buy-and-Hold portfolio.

- Recovery Behavior: Portfolio WF 7/28 lost more than Portfolio WF 14/10 during Buy-and-Hold recovery phases.
- Overall Performance: For about 70% of the time in the Unseen Data Period, the WF portfolios' strategies outperformed the Buy-and-Hold portfolio, but ultimately, the Buy-and-Hold portfolio achieved the highest overall return.



Note: Equity curves for portfolios equal weights at inception: 1. Buy-and-hold portfolio – BTC, ETH, BNB equally weighted. 2. Portfolio – walk-forward strategy with training 7 days and testing 28 days – strategy applied to BTC, ETH, BNB equally weighted. 3. Portfolio – walk-forward strategy with training 14 days and testing 10 days – applied to BTC, ETH, BNB equally weighted. No rebalancing during the whole period. The unseen data period covers data from November 7, 2019, to August 22, 2021. Each transaction incurs a 0.1% cost. Changing positions from short to long requires two transactions, resulting in a total cost of 0.2%

7 Summary

We divided the data into roughly equal halves: a global training period and an unseen period, each spanning over 18 months of intraday data. Instead of optimizing EMA parameters directly, we used a novel walk-forward process to optimize training and testing window lengths. This involved multiple walk-forward optimizations on the global training period to find combinations maximizing risk-adjusted returns via the Robust Sharpe Ratio.

To avoid cherry-picking bias, unseen data was used only once without further optimizations. Two optimal walk-forward parameter sets from training were applied to the unseen period. These strategies slightly underperformed Buy-and-Hold in the Sharpe Ratio but outperformed in maximum drawdown. Results were mixed for Sortino and Information Ratios: one set outperformed Buy-and-Hold, the other lagged. Notably, the top training performer underperformed in the unseen period.

Strategy robustness was validated by applying the same parameters (optimized on Bitcoin) to Ethereum and Binance Coin in the unseen period—assets unused during optimization. Results

mirrored Bitcoin: Buy-and-Hold outperformed both walk-forward strategies more in the Sharpe Ratio; for other metrics, outcomes were mixed. The 14/10-day training/testing strategy beat Buy-and-Hold in Information Ratio and maximum drawdown for ETH, but for BNB, Buy-and-Hold excelled in most metrics except drawdown.

Overall, the top training parameters (7/28 days) underperformed Buy-and-Hold and the 14/10-day alternative in the unseen period, questioning the parameter selection method’s optimality.

Sharpe Ratio performance dropped 20% on average from training (top 2 of 81 combinations) to testing; the best combination fell 30%, the second 10%.

From unseen period results across three cryptocurrencies, we built portfolios using: (1) walk-forward 7/28 days, (2) walk-forward 14/10 days, and (3) Buy-and-Hold. Both walk-forward portfolios showed lower drawdown (7/28: 10% less; 14/10: 20% less) and volatility than Buy-and-Hold. Buy-and-Hold led in Sharpe Ratio, but the 14/10 walk-forward beat it in Information Ratio, while 7/28 lagged 20%. Equity curves revealed Buy-and-Hold gains surged in the unseen period’s second half, with walk-forward leading initially.

Findings show the EMA strategy excelled in high-volatility periods with large asset gains/losses, surpassing Buy-and-Hold, but underperformed in low-volatility phases. It consistently reduced maximum drawdown versus Buy-and-Hold.

Heuristically, equity curves indicate our strategy outperformed Buy-and-Hold over a 14-month unseen period. Under a common 2:1 training/testing ratio ending after a year, it would still excel. This prompts questions on parameter validity, duration, and retraining timing. All 60-minute frequency training combinations outperformed Buy-and-Hold, with Sharpe Ratios above 0 versus Buy-and-Hold’s.

The hybrid portfolio—combining Buy-and-Hold with two walk-forward portfolios—outperformed all individuals, single assets, and portfolios across metrics, highlighting diversification value even without rebalancing.

Finally, against random strategies via two custom bootstrap methods, one showed statistical significance for both top parameter combinations.

8 Conclusions

The development of trading strategies in stock and cryptocurrency markets has been extensively explored through technical analysis in prior research. However, these studies frequently overlook critical elements such as transaction fees, slippage, and other trading costs. Additionally, a common but often unreported issue involves repeated optimizations on data designated as out-of-sample or the complete omission of an out-of-sample evaluation phase. Such practices foster unrealistic performance expectations, leading to substantial underperformance when strategies are implemented in live trading environments.

This study aimed to develop a trading strategy combining the Exponential Moving Average (EMA) and walk-forward optimization, assessing its performance on nearly four years of intraday

cryptocurrency data. Rather than maximizing results through EMA parameter optimization, a unique approach tested how data length periods influence strategy performance. Results show that specific training/testing period lengths in walk-forward optimization yield more favorable outcomes, contradicting the Efficient Market Hypothesis, which implies no differences across period lengths.

The study addressed the following questions:

RH1: What is the optimal length of training/testing periods for the walk-forward technique and moving average crossover strategy? The investigation sought ideal training and testing lengths. Initial results favored 7-day training and 28-day testing, but unseen data showed a 14-day training and 10-day testing combination as superior. Periods longer than 7 days for both may generally be advantageous.

RH2: How do the choice of data frequency and costs impact strategy results? Performance was evaluated across timeframes from 1 to 60 minutes. With 0.1% transaction costs, only intervals over 30 minutes appeared profitable, as shorter ones fail to overcome costs. Below 60 minutes should be avoided at this cost level. Breakeven was around 0.4% per transaction for 60-minute timeframes.

RH3: Is the tested strategy better than random? The strategy was compared to random approaches via two bootstrapping methods. Block bootstrap indicated superiority at 5% confidence. Random EMA bootstrap found only one random strategy slightly better, affirming merit.

Walk-forward optimization prevented overfitting in parameter estimation. The study examined varying training and testing lengths' effects, enhancing efficacy and theoretical understanding. Data were partitioned into global training for optimization and unseen periods for execution, minimizing bias—unlike iterative out-of-sample testing. Intraday data for Bitcoin, Ethereum, and Binance Coin were used, with Bitcoin parameters applied across assets. Limitations include order book impact (average bid-ask prices may not reflect large-order executions), bid-ask uncertainty (0.1% cost optimistic for illiquid exchanges), holding costs (for futures), shorting availability (assumed but variable), data splitting effects, and unexplored biases.

The novelty lies in examining walk-forward window lengths, revealing performance impacts beyond arbitrary values, and opening theoretical frameworks. Findings advise dynamic period lengths and single out-of-sample evaluations. The approach reduced drawdown and improved risk-adjusted metrics, allowing equivalent profits with lower risk, enhancing stability, confidence, and crash resistance. EMA excelled in high volatility but lagged in low volatility; all 60-minute training combinations outperformed Buy-and-Hold (Sharpe >0).

The combination of Exponential Moving Average and walk-forward optimization merits further investigation. Avenues include temporal parameter validity, exploring re-evaluation every 2-3 months or varying lengths for improvement, backtesting with updated data, alternative EMA selection minimizing training/testing deviation for robustness, risk management beyond full position, and mean reversion with dynamic switching for recovery underperformance.

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